

wev: Distilling LLM Browser Agents into Open, Local System-One Decision Models

AUTHOR NAME
AFFILIATION
EMAIL

Abstract

System-One decision models answer typed questions (a choice, a yes/no, a score) about a free-form state with a probability for every option, in one forward pass and without generating text. Browser agents make such decisions at every step (*which operation? which element?*), yet the open System-One models released so far are trained on general classification and rule data. We show that they do not transfer to browser steps: on Mind2Web test requests rendered in the request format of an open browser agent, Kev-4B and Kev-8B get 21% and 19% of steps fully right and Laya under 1%. We release **wev**, a family of open decision models (1.7B, 4B, 8B) built on Qwen3 base models, trained on converted Mind2Web and NNetNav steps, general typed-decision corpora, and episodes collected from a prompted LLM acting as the System One of a browser agent on live websites, verified by an LLM judge. wev-4b gets 75.9% of Mind2Web test steps right while matching or exceeding Laya on general typed-decision benchmarks; it trails Kev by 6–8 points on Kev’s own suites. Used as the System One of the same agent on 153 held-out live-website tasks, wev-4b completes 30 tasks and wev-8b 28, on par with the qwen3-max teacher (27), at 15–320 ms per decision on one consumer GPU. We also find that about a third of NNetNav’s DONE labels do not hold up under an LLM judge. Code, data builders, models and all evaluation outputs are public.

1 Introduction

Agents spend most of their steps on bounded decisions rather than on writing: which tool is next, whether a page shows the answer, which element to click. System-One decision models [TypeSafe AI, 2026] target exactly this. The caller declares questions and their options at request time; the model returns a probability for every option, with nothing to parse and nothing that can be malformed. Since the release of the hosted Jev model, open reproductions have appeared [Palmer, 2026, M, 2026, Cheng et al., 2026], together with studies of how to use such models inside agents [Wu and Lim, 2026] and of their failure modes [Sun and Xu, 2026].

Browser agents are a natural client. The open agent jev-ultrafast [Browser Use, 2026] already asks a hosted System One, at every step, which operation to perform and which listed element to target. Yet every open decision model we know of is trained on general decision data. A browser step differs from those data in two ways: the state is long (a page of text and 8–40 candidate elements, about 3,200 tokens in our data), and the right answer depends on the page rather than on world knowledge or a rubric.

This report asks a practical question: *can an open, local decision model serve as the System One of a browser agent, and what does it take to train one?* Our contributions are:

- **Evidence that general decision models do not transfer to browser steps.** Scored on the same requests with the same rule, Kev and Laya get at most 21% of Mind2Web test steps right (§7.2).
- **Data.** Converters from Mind2Web [Deng et al., 2023] and NNetNav [Murty et al., 2024] to the agent’s exact request format; an LLM-judge relabelling of NNetNav’s stopping decisions, which overturns 31% of its DONE labels; and a pipeline that collects judge-verified episodes from a prompted LLM running the agent on live sites (§4).
- **Models.** wev-1.7b, wev-4b and wev-8b: 76% step success on Mind2Web test with general typed-decision accuracy at or above Laya’s, and live-website task success on par with the LLM teacher at 15–320 ms per decision (§7).
- **A reproducible evaluation.** One scorer for every model, per-request results, and an end-to-end harness on live websites with judge-verified success. Everything is released.

We do not propose a new architecture: the model follows Kev’s design (§3), and two structural variants we tried did not help at scale (§7.5).

2 Related work

System-One decision models. Jev [TypeSafe AI, 2026] introduced the hosted model class; its training method (RLCD) is unpublished. Kev [Palmer, 2026] attaches LoRA adapters and a pointer head to Qwen base models and trains on ten public classification sources plus rule records; Laya [M, 2026] fine-tunes ModernBERT-large [Warner et al., 2024] with a small decision head; this-that-model [Cheng et al., 2026] reads a 2B model’s answer from a designated position. REFLEX [Wu and Lim, 2026] places hosted Jev inside an LLM agent and escalates to a strong LLM when confidence is low; Sun and Xu [Sun and Xu, 2026] show that decision heads can follow the option name rather than the rubric bound to it. None of these trains or evaluates a decision model on browser steps.

Web agents. Mind2Web [Deng et al., 2023] provides human demonstrations on real websites; NNetNav [Murty et al., 2024] collects live-web trajectories by environment interaction. Most web agents generate actions with an LLM. We instead cast each step as two typed questions and distil a generative teacher into a non-generative student, in the spirit of knowledge distillation [Hinton et al., 2015] and imitation learning [Ross et al., 2011].

3 Model

We follow Kev’s design [Palmer, 2026] and adapt its code (Apache-2.0).

Rendering. A request is a state and a set of questions. The state is written once; each question follows it as its own branch, with its options delimited by reserved tokens of the Qwen vocabulary:

```

    <state> page text · elements · recent actions
    <q> which operation? <opt> CLICK </opt> <opt> TYPE_TEXT </opt> ... <decide>
    <q> which element to CLICK? <opt> [1] Search </opt> <opt> [2] Sign in </opt> ... <decide>

```

Delimiter strings that appear in user text are rewritten, so text cannot forge structure.

Mask and positions. Every token of a question branch attends to the state and to its own branch, never to another question; each branch restarts its positions after the state. One forward pass therefore answers all questions of a request, and neither question order nor packing changes an answer (verified in tests to fp32 noise).

source	license	train	weight	role
Mind2Web [Deng et al., 2023]	CC BY 4.0	5,863	1	human browser steps, split by website
NNetNav-live [Murty et al., 2024], relabelled	Apache-2.0	11,408	1	live-web steps incl. DONE, BLOCKED, SCROLL
teacher episodes, first collection	qwen3-max outputs	2,548	3	judge-verified live-web episodes
teacher episodes, second collection [†]	qwen3-max outputs	2,274	3	tasks the first collection failed
kev decision-v7 [Palmer, 2026]	per source	12,576	2	classification, QA, rule records
typed-decisions (80% of train)	Apache-2.0	960	8	agent/ops workflows, 5 questions per case
tasksource-jev	mixed, some research-only	57,024	1	392 classification tasks as decisions
jev-distill-corpus-v3	Apache-2.0	50,000	1	synthetic scenarios, soft labels
typed-decisions-synth	MIT	6,682	1	multi-question cases, 149 domains

Table 1: Training mixture (rows after conversion; weight = repetitions per epoch). [†]wev-4b only.

Backbone and readout. A Qwen3 base model [Qwen Team, 2025] with its vocabulary head removed (it cannot generate), adapted with LoRA [Hu et al., 2022] (rank 16 on every attention and MLP projection). A pointer head projects the hidden state of each option’s `</opt>` token and of the question’s `<decide>` token to 256 dimensions and scores their dot product; a softmax over the options gives the answer. The adapter is merged into the released weights.

Long states. Browser states can exceed the context used in training. At inference the page text, URL, per-element option lists and action history are shortened in that order until the request fits; a state of up to 4,096 tokens and question branches of up to 8,192 tokens are accepted.

4 Data

Table 1 lists the training mixture. All browser sources are rendered exactly as jev-ultrafast renders its per-step request (state fields, operation options, instruction text, and `[index]` `label` targets).

Mind2Web. Each recorded step becomes one request: the page’s text and a sample of 8–40 candidate elements (the gold element plus negatives), the task, and the preceding actions; the questions ask for the operation and, for that operation, the target. Splits are by website, so test sites are unseen in training (train 5,863, dev 586, test 875 requests; 873 after removing two whose element lists exceed the API’s 255-option limit).

NNetNav and its stopping labels. NNetNav supplies the operations Mind2Web lacks: stopping (DONE), giving up (BLOCKED) and scrolling. Its labels come from unsupervised exploration, and stopping labels are noisy. We asked an LLM judge (DeepSeek-V4.1-Flash), given the goal and the page at each step, whether the goal was already achieved. On the training split, the judge rejected 592 of the 1,898 DONE labels (31%) and found the goal already achieved at 2,281 of the 10,102 other steps (23%). We relabel the latter as DONE and drop the former, since the correct next step is unknown.

Teacher episodes. We serve a prompted LLM (qwen3-max) behind the System One API and let jev-ultrafast use it on live websites, logging every request and answer. Tasks (1,402 train, 153 held-out evaluation) come from NNetNav goals on their live sites, Wikipedia look-ups and Google Flights searches. We keep every decision of an episode that ended in DONE when a judge, reading the final page, confirms the goal; every decision up to a BLOCKED on a page that blocked the agent (CAPTCHA, 403, login wall); and, for episodes cut short by the browser infrastructure, every decision but the last. Episodes that loop or exhaust their budget are dropped. Of 164 DONE claims in the second collection, the judge confirmed 103. Train and dev are split by task.

We found two harness failures that inflated error rates in early collections: each episode left a browser daemon running (thousands accumulated and starved the machine), and a page read during navigation aborted the episode. Both are fixed in the released collector, and all end-to-end numbers below use the fixed harness.

General decisions. Kev’s decision-v7 training split, 80% of the typed-decisions train split, and three public corpora (tasksource-jev, jev-distill-corpus-v3, typed-decisions-synth), using soft labels where provided. Rows whose state matches a typed-decisions test state are removed.

5 Training

One epoch, AdamW with learning rate 10^{-4} and a one-cycle schedule (10% warm-up), cross-entropy against the (soft) label distribution of every question, frozen bf16 base weights with fp32 adapters and head. Requests are packed into length-bucketed micro-batches of at most 6,000 padded tokens per GPU for wev-4b and wev-8b, which train data-parallel on 8 NVIDIA H20 GPUs (66 and 89 minutes), and of 20,000 tokens for wev-1.7b on one GPU (146 minutes).

6 Evaluation

Benchmarks. Kev decision-v7 test (1,176 requests) is Kev’s in-domain suite; Kev transfer-v4 test (764) is out-of-domain for every model; typed-decisions test (400 cases, 5 questions each) covers four agent/ops workflows; Mind2Web test (873) and NNetNav test (1,150) are browser steps.

Scoring. Every model receives identical requests; its top option is its answer. We report per-question accuracy for general benchmarks and *step success* (operation and target both right) for browser steps. Kev runs through its own server (bf16) and Laya through its Python package, both as published. Requests beyond wev’s evaluation context (11 of 873 on Mind2Web) count as wrong for wev.

End to end. jev-ultrafast runs each of the 153 held-out tasks with the model as its System One; success means the agent stops with DONE and a judge, reading the final page, agrees.

Test protocol. Test splits were held out from training and model selection, with one exception: wev-4b and wev-8b each had two final candidates (with and without the second teacher collection), and both were read on test. The choice for wev-4b rests on development and end-to-end results; for wev-8b it was made after both test reads.

7 Results

7.1 General typed decisions

Table 2. wev matches or exceeds both Laya models on every general benchmark. On Kev’s own suites, Kev leads: by 6–8 points on decision-v7, whose training split both train on, and on transfer-v4 Kev-4B leads by 8 points over wev-4b while wev-8b is level with Kev-8B, which shares its base model. The typed-decisions column favours models trained on that dataset: an earlier 4B wev trained without it (and without the external corpora) scored 53.1% on its development split. Our Laya (typed-decisions) score reproduces its published 76.6%.

model	decision-v7	transfer-v4	typed-decisions
wev-1.7b	81.1	65.5	79.5
wev-4b	80.6	73.8	79.4
wev-8b	82.4	77.2	79.1
Kev-4B (Qwen3.5-4B)	88.2	82.1	65.1
Kev-8B (Qwen3-8B)	88.1	76.8	62.7
Laya (typed-decisions)	65.7	62.8	76.8
Laya	64.3	63.7	36.2

Table 2: General typed decisions, test splits, per-question accuracy (%). wev and Laya (typed-decisions) train on the typed-decisions train split; Kev and Laya do not.

model	Mind2Web test		NNetNav test		
	step	operation	step	DONE recall	premature DONE
wev-1.7b	68.2	88.1	61.0	80.4	9.4
wev-4b	75.9	91.2	61.4	84.6	10.0
wev-8b	75.5	90.3	60.8	79.6	8.2
Kev-4B	21.2	35.7			
Kev-8B	19.0	73.3			
Laya (typed-decisions)	0.7	13.1			
Laya	0.0	2.5			

Table 3: Browser steps (%). Step: operation and target both right; targets are chosen among 8–40 candidates. Premature DONE: share of not-finished steps answered DONE.

7.2 Browser steps

Table 3. General decision models fail on browser steps as published: Kev-8B picks the right operation on 73% of steps but the right element far less often, and Laya’s context (512–1,024 tokens) holds only the start of a page. wev gets three quarters of Mind2Web test steps right on websites it never saw. On NNetNav the hardest decision is when to stop: wev recalls 80–85% of DONE steps and says DONE too early on 8–10% of the others; thresholding the DONE probability trades one for the other.

7.3 End to end on live websites

Table 4. The local students complete as many tasks as their teacher. The differences are within run-to-run noise: two runs of wev-8b variants with equal totals disagreed on 8 tasks in each direction. Google Flights often answers automated browsing with a CAPTCHA, and every model fails most of those tasks.

7.4 Latency

Table 5. All questions of a request share one forward pass, so five questions cost little more than one; browser steps are dominated by encoding the page. On a 16 GB Apple M5 laptop (PyTorch MPS backend, fp32), wev-1.7b loads in about 30 s and answers a one-question decision in 77 ms.

System One	all (153)	NNetNav goals (123)	Wikipedia (10)	Google Flights (20)
qwen3-max, prompted (teacher)	27	13	10	4
wev-4b	30	18	10	2
wev-8b	28	18	10	0

Table 4: Judge-verified task successes on held-out live-website tasks with jev-ultrafast.

model	decision-v7	typed-decisions	browser step
	1 q., ~180 tok.	5 q., ~380 tok.	2 q., ~3,200 tok.
wev-1.7b	10 ms	13 ms	91 ms
wev-4b	15 ms	25 ms	217 ms
wev-8b	21 ms	37 ms	322 ms

Table 5: Median in-process latency, one RTX 5090, bf16, one request at a time.

7.5 What did not help

Structural variants. We tried two changes to the architecture on 1.7B (three seeds each, an earlier data mix, development splits): a set head, which runs a two-layer permutation-equivariant transformer over the option states before scoring, and truncating the backbone to its first 18 of 28 layers.

1.7B variant	Mind2Web dev step	NNetNav dev step
pointer head, 28 layers	69.0 $\pm$ 2.7	55.2
set head, 28 layers	66.5	53.9
set head, 18 layers	68.5 $\pm$ 0.2	56.6

The truncated model ran 23% faster and was more stable across seeds, but a later same-hardware control run put its cost at about 3 points, and the set head did not help at full depth. The released models use the pointer head and every layer.

A second epoch. Training wev-4b for two epochs changed development accuracy by less than one point on every benchmark (transfer-v4 74.6 $\rightarrow$ 75.1, decision-v7 81.9 $\rightarrow$ 82.6, typed-decisions 79.7 $\rightarrow$ 79.2, Mind2Web step 80.0 $\rightarrow$ 79.3) at twice the cost.

The second teacher collection. Adding episodes from tasks the first collection failed raised wev-4b’s live-website successes from 21 to 30; per task, 10 were gained and 1 lost. Its test accuracy was unchanged (within 0.6 points everywhere). On wev-8b the same data left live-website successes at 28 (8 gained, 8 lost) and lowered transfer-v4 by 1.8 points, so wev-8b keeps the earlier mix.

8 Limitations

- **Evaluation size.** 153 live tasks give wide intervals, and live sites change between runs; we report paired comparisons where we compare versions.
- **Candidates.** Targets are chosen among the elements the agent lists (8–40 per step), not every element on the page; the numbers are not comparable to the Mind2Web leaderboard.
- **Test reads.** See the test protocol in §5; wev-8b was chosen after two test reads.

- **Scope.** English only; no comparison with hosted Jev (no API access); decisions only, with typed text values produced by the agent’s separate text model.
- **Use terms.** Some training data is research-only (parts of tasksource-jev) or subject to its provider’s terms (qwen3-max outputs). We release the models as research artifacts.

9 Conclusion

General-purpose open decision models do not transfer to browser steps as published, but a decision model trained on converted demonstrations, relabelled live-web data and judge-verified teacher episodes does: wev gets three quarters of unseen-site steps right and, as the System One of an open browser agent, completes as many live tasks as the LLM it was distilled from, locally and in tens to hundreds of milliseconds per decision. Natural next steps are on-policy distillation, where the teacher labels the states the student reaches, and a larger live-task suite for tighter end-to-end comparisons.

Release. Code, data builders and evaluation outputs: <https://github.com/alanhuangyoo/wev>. Models: <https://huggingface.co/alanhuangya>. Package: `pip install wev-ai`.

References

- Browser Use. jev-ultrafast. <https://github.com/browser-use/jev-ultrafast>, 2026.
- Zehua Cheng, Wei Dai, and Jiahao Sun. this-that-model-1.0: A typed decision model that decides in 30 ms, for a millionth of a cent. *arXiv preprint arXiv:2609.23886*, 2026.
- Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2web: Towards a generalist agent for the web. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2023.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Nandha Kishor M. Laya: a non-autoregressive system 1 decision engine. <https://github.com/NandhaKishorM/laya>, 2026.
- Shikhar Murty, Hao Zhu, Dzmitry Bahdanau, and Christopher D. Manning. NNetNav: Unsupervised learning of browser agents through environment interaction in the wild. *arXiv preprint arXiv:2410.02907*, 2024.
- Jared Palmer. kev: open decision models on qwen base models. <https://github.com/jaredpalmer/kev>, 2026.
- Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the International Conference on Artificial Intelligence and Statistics*, 2011.

Yu Sun and Junhao Xu. Type-safe is not error-free: A constrained decision head follows the option name, not the rubric bound to it. *arXiv preprint arXiv:2609.26758*, 2026.

TypeSafe AI. Jev: a system one model for typed, calibrated decisions. <https://typesafe.ai>, 2026.

Benjamin Warner et al. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*, 2024.

Tiantong Wu and Wei Yang Bryan Lim. REFLEX with jev for efficient selective control in LLM agents. *arXiv preprint arXiv:2609.26532*, 2026.